Recommended Course List

Quantitative Trading & Investment Association

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About This Document:

- This document has been compiled by QTIA to help CUHK-Shenzhen students interested in quantitative finance make informed course selections and develop a more coherent academic plan. It is intended for reference only and should not be interpreted as the graduation requirements of any academic program.
- Courses are organized by knowledge blocks. Within each block, courses are generally ordered from more introductory to more advanced, and students are advised to take courses within the same block in the listed order when possible. Highlighted and underlined block titles indicate areas that should be prioritized.
- This document draws in part on course guides and recommendations prepared by previous QTIA members. The current editor has not taken every listed course, and course content, instructors, and offering patterns may change over time. Detailed advice therefore cannot be provided for every course. Please evaluate the recommendations carefully and make decisions according to your own academic plan.
- The University's official study plans for the 2026–2027 academic year include several newly introduced courses. Based on their published course content, the editor has included selected courses that appear particularly valuable, although the editor has not taken them. These courses are highlighted in blue and should be considered with appropriate discretion.
- We thank experienced industry professionals for reviewing this study plan. We also thank previous QTIA members and graduates for preparing its earlier versions and providing valuable suggestions for subsequent revisions.
- Questions, corrections, and suggestions regarding this document are welcome. Please contact us at st_qtia@link.cuhk.edu.cn.

1 Finance & Economics

Code[1]	Course Title	Offered[2]	Note[3]
Economics			
ECO2011	Basic Microeconomics	Spring&Fall	
ECO2021	Basic Macroeconomics	Spring&Fall	
ECO3121	Introductory Econometrics	Spring&Fall	
Derivatives & Asset Pricing			
FIN2010	Financial Management	Spring&Fall	
FIN4110	Options and Futures	Spring&Fall	
FIN4120	Fixed Income Securities Analysis	Fall	Spring Version: FIN6116 FIN6116 is equivalent to FIN4120 in course content.
FIN4231	Asset Pricing	Spring	
FIN6203	Asset Pricing	Fall	
Portfolio Management & Quantitative Investing			
FIN3080	Investment Analysis and Portfolio Management	Spring&Fall	
DDA3600	Factor Investing	Spring	See Prof. Chuan SHI's course page for more details.
FIN6106	Quantitative Portfolio Analysis	Fall	
FIN6131	Portfolio and Asset Management	Spring&Fall	FIN6131 in **Fall**[4] and MFE5340 cover most content of DDA3600, with further extensions. See Prof. Chuan SHI's teaching page for details.
MFE5340[5]	Artificial Intelligence in Financial Engineering: Quantitative Investment	Spring&Fall	
Algorithmic Trading			
FIN6115	Microstructure and Algorithm Trading	Fall	FIN6115 focuses on foundational concepts; MFE5210 is more industry-oriented; MFM5220 is more theoretical in financial mathematics.
MFE5210[5]	Algorithm Trading	Spring	
MFM5220	Algorithm Investment	Spring	
FinTech & AI Applications			
FIN3210	Fintech Theory and Practice	Fall	
FIN3090	AI Applications in Finance	Fall	

[1]For equivalent or cross-listed courses, the table lists the recommended or most appropriate code; near-duplicate graduate-level versions are not repeated unless the course is materially different. Highlighted in yellow and underlined course codes indicate core courses strongly recommended for students interested in quantitative finance or trading.

[2]Offered terms are based on recent records and may be inaccurate. Courses not offered in the past two academic years are generally excluded; future offerings may change. Please verify historical offerings in SIS when needed.

[3]Notes list related versions, tracks, or alternative course codes when applicable.

[4]This course will no longer be taught by Prof. Chuan SHI in future offerings.

[5]MFE courses are offered by the Master of Finance program and are not open for undergraduate enrollment; undergraduate students may attend only as auditors.

Code	Course Title	Offered	Note
FIN3380	Financial Data Analysis with AI Tools	Fall	

2 Mathematics & Statistics

Code	Course Title	Offered	Note
Linear Algebra & Mathematical Analysis			
MAT2040	Linear Algebra	Spring&Fall Summer	Honours Version: MAT2042; SDS Version: MAT2041 or MAT2041A[6]
MAT1001	Calculus I	Fall	Honours Version Track: MAT1011, MAT1012, MAT2060[7]
MAT1002	Calculus II	Spring	
MAT2050	Mathematical Analysis	Fall	
MAT3006	Real Analysis	Fall	
Probability Theory & Stochastic Processes			
STA2001	Probability and Statistics I	Spring&Fall Summer	Honours Version: STA2001H Strongly recommended for students planning to take MAT3280.
MAT3280	Probability Theory	Spring	Graduate Version: DDA6020[8]
STA4001	Stochastic Processes	Spring&Fall	Honours Version: STA4001H
CIE6005	Stochastic Process	Spring	Recommended Prerequisite: MAT3006
DDA6001	Advanced Stochastic Process	Spring	
DDA4002	Stochastic Simulation	Fall	
DDA6040	Dynamic Programming and Stochastic Control	Spring&Fall	
Optimization			
MAT3007	Optimization	Spring&Fall Summer	Honours Version: MAT3007H Recommended for in-depth optimization.
MAT3220	Optimization II	Spring	Fall Version: DDA6010 DDA6010 is equivalent to MAT3220 in course content.
DDA6110	Advanced Convex Optimization	Spring	These are solid options; interested students may choose one or two.
DDA6112	Stochastic Optimization	Fall	
DDA6207	Advanced Optimization	Fall	

[6] For most students interested in quantitative finance or trading, MAT2041 or MAT2041A may be more directly applicable than MAT2042. MAT2042 is more proof-oriented and may be more suitable for students with a strong interest in theoretical mathematics.

[7] Students planning to take MAT3006 are strongly encouraged to take the honours track; otherwise, MAT1001, MAT1002, and MAT2050 are strongly recommended.

[8] MAT3280 assumes a solid MAT2050 or MAT2060 analysis background. DDA6020 may be considered as a graduate version of MAT3280 for selected students; it is substantially harder, and MAT3006 is strongly recommended beforehand.

Code	Course Title	Offered	Note
Statistical Inference & Modeling			
STA2002	Probability and Statistics II	Spring&Fall Summer	Honours Version: STA2002H Strongly recommended for students planning to take STA3020.
STA3001	Linear Models	Fall	
STA4102	Generalized Linear Models	Spring	
STA4002	Multivariate Statistical Analysis	Spring	
DDA4010	Bayesian Statistics	Spring	
STA3020	Statistical Inference	Spring	
Differential Equations			
MAT2002	Ordinary Differential Equations	Spring	Honours Version: MAT2001[9]
MAT4220	Partial Differential Equations	Spring	
MAT4240	Numerical Methods for Differential Equations	Fall	
MAT4500	Stochastic Differential Equations	Fall	
Numerical Analysis & Stochastic Numerical Methods			
DDA3005	Numerical Methods	Fall	Mathematical Version: MAT4001 MAT4001 assumes a stronger mathematical background.
MAT3240	Numerical Linear Algebra	Spring	
CSC4220	Matrix Computation	Fall	PhD-level course open to undergraduates[10] Recommended Prerequisites: MAT2041 or MAT2041A, DDA3005
STA4504	Advanced Monte Carlo Methods	Fall	
Time Series			
STA4003	Time Series	Fall	
Financial Mathematics			
MAT6810	Advanced Financial Models	Fall	
MAT7710	Topics in Financial Mathematics	Spring	
Financial Statistics			
FMA4200	Financial Data Analysis	Spring	
FMA4800	Financial Computation	Spring	Recommended Prerequisite: FIN4110
STA4020	Statistical Modeling in Financial Markets	Fall	
MFM5130	Statistical Methods for Risk Modeling	Fall	

[9]For most quant-oriented students, MAT2002 is usually sufficient; MAT2001 is more proof-oriented.

[10]This course covers numerical linear algebra, an important topic in numerical methods; it is difficult and requires a strong linear algebra background.

3 Computer Science & Data Science

Code	Course Title	Offered	Note
Programming & Computer Systems			
CSC1001	Introduction to Computer Science: Programming Methodology	Fall	SSE Version Track: CSC1005, CSC1006 SDS Version Track: CSC1003, CSC1004
CSC1002	Computational Laboratory	Spring	
CSC3002	C/C++ Programming	Spring&Fall	
CSC3050	Computer Architecture	Spring	
CSC3150	Operating System	Spring&Fall	
CSC4005	Parallel Programming	Fall	PhD-level course open to undergraduates
Machine Learning			
DDA3020	Machine Learning	Spring&Fall	Honours Version: DDA3020H
DDA4210	Advanced Machine Learning	Spring	
DDA4220	Deep Learning and Applications	Fall	
DDA4230	Reinforcement Learning	Spring	
CSC4100	Natural Language Processing	Spring	
DDA6201	Selected Topics in Data and Decision Analytics I	Fall	
DDA6202	Optimization Models and Methods in Machine Learning	Fall	
Algorithms			
CSC3001	Discrete Mathematics	Spring&Fall	
CSC3100	Data Structures	Spring&Fall Summer	
CSC4120	Design and Analysis of Algorithms	Spring&Fall	Recommended Prerequisites: CSC3001 **and** CSC3100 Honours Version: CSC4120H
Databases			
CSC3170	Database System	Spring&Fall	

Appendix A: Knowledge Mapping[11]

This map follows the end-to-end quantitative investment process. Within each module, knowledge blocks are listed roughly from more foundational to more advanced.

Data Warehouse →	Return Model →	Risk Model →	Portfolio Optimization →	Algorithmic Trading
Programming	Linear Algebra	Linear Algebra	Linear Algebra	Programming
Computer Systems	Economics	Mathematical Analysis	Mathematical Analysis	Algorithms
Databases	Statistical Inference	Probability Theory	Optimization	Computer Systems
Financial Statistics	Time Series	Statistical Inference	Numerical Analysis	Numerical Analysis
	Machine Learning	Time Series	Portfolio Management	Machine Learning
	Quantitative Investing	Stochastic Processes	Asset Pricing	Algorithmic Trading
	Asset Pricing	Differential Equations	Machine Learning	
		Financial Statistics		
		Derivatives		

Appendix B: Suggested Study Plan

This schedule reflects the editor's current view of a suitable course-taking pattern for students interested in quantitative finance. It lists only selected foundational courses, and students are encouraged to broaden or deepen their study based on their own academic background and career plans.

Year	Recommended Courses[12]	Note
Year 1	CSC1001, MAT1001	These courses are recommended because they are important prerequisites for later study.
	CSC1002, MAT1002, MAT2040, STA2001	
Year 2	CSC3100, ECO3121, STA2002 (FIN3080, MAT2050, MAT3007)	The six non-optional courses broadly cover in-class content relevant to factor-research-related summer internship interviews. The optional courses are useful for later study; whether to take them depends on each student's academic progress and internship situation.
	DDA3020, DDA3600, STA4001 (CSC3002, FIN4110, MAT2002)	
Year 3	CSC3002, DDA3005, FIN4120, MAT3007, STA4020	Since Year 3 is an important period for internships and exchange programs, students are not advised to take too many courses.
	DDA4080[13], FMA4800, MAT2002	
Year 4		Students may choose courses based on their own academic plans and on whether they want to broaden or deepen the knowledge blocks listed above. They should also complete courses required for their own major; therefore, no specific recommendations are provided here.

[11] This map is adapted from Prof. Chuan SHI's Curriculum Mapping for end-to-end quantitative investment processes.

[12] Optional courses are enclosed in parentheses.

[13] DDA4080 Capstone Project is open only to SDS students. It is an SDS school-enterprise collaboration course in which students work on projects with industry mentors; quantitative projects may be available. For details, see the official WeChat article.